When Do Quantum Models Help? A Cross-Domain Empirical Study

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***Abstract*—While hybrid quantum-classical neural networks theoretically offer rich representational capacity in high-dimensional Hilbert spaces, empirical benchmarks frequently fail to isolate the quantum contribution from classical artifacting. Consistent with recent literature, we find that quantum models do not systematically outperform classical baselines across all regimes. To identify the specific conditions under which quantum models provide measurable utility, this paper introduces a strict 1-to-1 parameter-ablated benchmark across a 2×2 empirical matrix of data topology (tabular vs. geometric) and volume (sufficient vs. extreme $p \gg N$ starvation), and develops QKAResQNet, a hybrid architecture coupling a classical residual highway with trainable Quantum Kernel Alignment through a learnable gating parameter. We identify four operational boundaries: (1) a geometric utility regime under data starvation, where quantum twins outperform parameter-matched classical twins on topologically complex manifolds (up to $+24.48$ percentage points), while remaining below the exact classical RBF-SVM on these geometric manifolds, where it acts as a reference ceiling (on two higher-dimensional sets a neural twin slightly exceeds it, and on quantum-native data the RBF-SVM is not applicable); (2) representational parity under multi-omics feature starvation, where a 4-qubit bottleneck matches classical network capacity with tighter variance; (3) standard-regime parity, where classical models dominate strictly due to training-time efficiency; and (4) a classical compression boundary, where hybrid architectures fail on quantum-native entangled data because classical compression destroys multipartite entanglement. We further evaluate QKAResQNet on a physical 156-qubit IBM Kingston processor, where hardware inference classifies 13 of 14 held-out test samples correctly, matching the RBF-SVM count on this small feasibility sample. These findings establish inductive bias alignment between the parameterized quantum circuit and the data manifold as a necessary condition for NISQ-era quantum utility.**

***Index Terms*—Quantum Machine Learning, Hybrid Neural Networks, QKAResQNet, Quantum Kernel Alignment, Data Starvation, Parameterized Quantum Circuits, Inductive Bias.**

I. Introduction

Parameterized quantum circuits (PQCs) have been proposed as a route to richer inductive bias and enhanced representational capacity [5]–[7]. However, the Noisy Intermediate-Scale Quantum (NISQ) era is plagued by an epistemological benchmarking crisis: a substantial fraction of papers claiming a "quantum advantage" rely on unfair comparisons, artificial datasets, or mismatched parameter counts [13]–[15]. In practice, it remains unclear exactly when hybrid quantum models genuinely provide utility over mature classical networks. In small-data regimes, especially when the number of features greatly exceeds the number of samples ($p \gg N$), classical models suffer from interpolative collapse, while quantum models often fall victim to barren plateaus or trap-swamped optimization landscapes [10]–[12]. Generalization from limited training data remains a fundamental challenge [1]–[3]. This work focuses on a targeted empirical question: *when does the quantum component actually help?* We adopt an explore-then-develop methodology. We first explore five quantum architectures across a 2×2 empirical matrix (topology vs. volume) under strict 1-to-1 parameter ablation, and then develop QKAResQNet, an architecture that couples a classical residual highway with trainable Quantum Kernel Alignment through a learnable gating parameter γ, designed to address the barren-plateau instability and static feature-collision failures exposed during exploration. We abandon the search for universal supremacy and instead map the true operational boundaries of hybrid NISQ-era models, evaluating the final architecture on physical quantum hardware.

II. Methodology

A. Architectural Design

We explore three hybrid topologies: **ResQNet**, **SplitAttentionQNN**, and the hero architecture **QKAResQNet**. ResQNet combines a classical compression stage with a small 4-qubit quantum bottleneck and a residual bypass connection, building upon recent lean hybrid topologies [9], [16]. The classical front-end maps the input into a low-dimensional latent space, while the quantum block processes that representation using a hardware-efficient parameterized quantum circuit. The residual path preserves gradient flow, bypassing barren plateaus. For high-dimensional topological inputs, SplitAttentionQNN spatially chunks the feature space, passing independent representations through parallel quantum embeddings combined by a learned attention layer. QKAResQNet retains the residual highway and dynamic gating of ResQNet but replaces the prototype's fixed compressor/feature map with a trainable, bias-free Quantum Kernel Alignment (QKA) layer at the input to the circuit, allowing the embedding geometry to adapt to each dataset during optimization. All three hybrids employ a learnable scalar gate γ that scales the quantum branch relative to the classical highway. The forward pass of QKAResQNet is:

$$\mathbf{y} = H(\mathbf{x}) + \gamma \cdot E\left(Q\left(\tanh(C_{\text{aligned}}(\mathbf{x}))\right)\right) \qquad (1)$$

where H is the classical highway, C_{aligned} the trainable, bias-free Quantum Kernel Alignment layer that maps inputs to the 4-qubit register (replacing the prototype's biased compressor), Q the parameterized quantum circuit (Z-feature-map encoding plus a RealAmplitudes ansatz), and E the classical expander. Initialized near zero, γ keeps the model classical early in training and increases as the classical path converges, progressively blending in the quantum contribution while maintaining stable gradients. Fig. 1 shows the data flow.

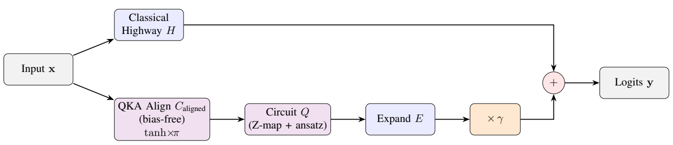

Fig. 1. The QKAResQNet architecture. A classical highway H runs in parallel with a quantum residual branch: a trainable, bias-free Quantum Kernel Alignment layer (C_{aligned}) maps the input to the qubit register, followed by Bloch phase scaling, the quantum circuit Q, and an expander E. The learnable gate γ scales the quantum branch before it is summed with the highway, yielding stable gradient flow and adaptive embedding geometry. The alignment layer is the single block changed relative to the prototype (ResQNet).

B. Unified 1-to-1 Ablation Strategy

To eliminate classical simulation artifacts, our experimental setup enforces a strict 1-to-1 replacement strategy. The classical and hybrid model twins share identical classical ingestion layers, scaling mechanisms, optimizers, learning rates, and total parameter counts. The sole variable altered is the central bottleneck: transitioning from a standard Euclidean dense layer to a unitary quantum circuit.

C. Statistical Protocol and Hardware Simulation

To accurately reflect physical NISQ constraints, the six core datasets were trained with density-matrix simulations incorporating the empirically derived single-qubit, two-qubit, and readout errors of the 133-qubit IBM Torino processor (Heron r1, 3.03% median readout error). Recognizing the extreme statistical instability of $p \gg N$ starvation regimes, model performance was evaluated across 20-seed randomized robustness sweeps, with each seed re-shuffling and re-splitting the data so that the reported variance reflects both the data partition and the model initialization. Statistical significance was assessed using both Welch's t-test and the Mann-Whitney U test to isolate genuine topological effects from initialization noise. For real-world evaluation, QKAResQNet was additionally deployed for inference on the physical 156-qubit IBM Kingston processor (Heron r2, 340K CLOPS); the gesture simulation results use 5-fold cross-validation over the full set, whereas the single physical-hardware run uses one stratified 75/25 train-test split.

III. Experiments and Results: Four Operational Boundaries

The empirical matrix spans geometric (Swiss Roll, Circles, Moons), chemical (QSAR), tabular (Breast Cancer), multi-omics (Leukemia, Colon), high-dimensional imaging (Brain RNA-seq, Indian Pines), and quantum-native (NTangled) datasets, with real-world ESP32 accelerometer gesture data used for physical-hardware evaluation. Table I summarizes the hero QKAResQNet pair against its parameter-matched CKAResCNet twin across all eleven datasets; the per-regime analyses follow.

TABLE I
QKAResQNet vs. Parameter-Matched Classical Twin (CKAResCNet) Across All Datasets (20-Seed Mean Accuracy, %)

Dataset	Classical	Quantum	Δ / Verdict
Swiss Roll	67.73	**82.16**	+14.42 Q wins
Circles	65.61	**76.50**	+10.89 Q wins
Moons	86.17	**91.26**	+5.09 Q wins
QSAR	82.21	**82.61**	+0.40 parity
Breast Cancer	**88.50**	85.14	−3.36 classical better
Leukemia	**70.85**	69.55	parity (p=0.70)
Colon	**66.26**	63.35	parity
Brain RNA-seq	**43.61**	43.11	parity (p=0.93)
Indian Pines	61.61	**61.94**	parity
NTangled	49.83	49.67	both fail (∼50)

A. The Topological Advantage (Geometric Starvation)

To evaluate inductive bias on complex continuous geometries under data starvation, models were tested on the Swiss Roll manifold with a 90/10 train-test split across 20 random seeds. As shown in Table II, the quantum twins demonstrate a consistent geometric utility under starvation. The *SplitAttentionQNN* reaches 91.77% accuracy, exceeding its classical twin by +24.48 percentage points with tighter variance, while the hero *QKAResQNet* contributes +14.42pp over its CKAResCNet twin ($p < 0.001$). The pattern holds across the other two synthetic manifolds: QKAResQNet beats its twin by +10.89pp on Circles and +5.09pp on Moons. The phase-encoded Hilbert space provides a stronger inductive bias for unrolling non-linear manifolds than flat Euclidean space.

TABLE II
Swiss Roll (20-Seed): Geometric Utility Under Starvation

Architecture Family	Classical Twin	Hybrid Quantum Twin
Split Attention	67.29% (±11.8%)	**91.77% (±4.4%)**
MLP Base	75.32% (±12.0%)	**84.44% (±9.3%)**
Residual (ResNet/ResQNet)	69.93% (±14.2%)	**82.56% (±7.8%)**
QKA Residual (CKA/QKAResQNet)	67.73% (±11.8%)	**82.16% (±9.2%)**
RBF-SVM (kernel ceiling)	98.89%	

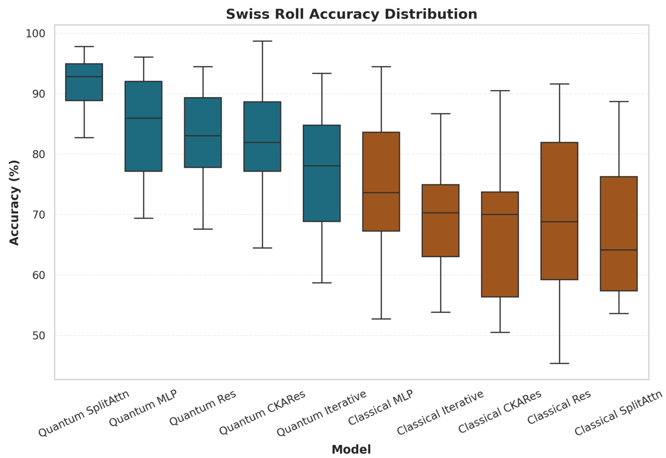

Fig. 2. Geometric utility on the Swiss Roll manifold (20-seed mean $\pm$ std). Every quantum twin outperforms its parameter-matched classical twin; the SplitAttentionQNN attains the largest margin (+24.48pp). All hybrids remain below the exact RBF-SVM ceiling of 98.89%, consistent with the kernel-approximation interpretation.

Crucially, on these geometric manifolds none of the hybrids exceed the exact RBF-SVM (98.89% on Swiss Roll), which serves as the reference ceiling in this regime; the hybrids approximate but do not surpass it here, and exact quantum kernels themselves face exponential concentration at scale [8]. We emphasise that this $O(N)$ kernel-approximation characterisation is asymptotic in sample count for the kernel evaluation and is *not* realised in wall-clock time at the present scale: as Section III-C shows, circuit simulation imposes a three- to four-order-of-magnitude time penalty over the classical twin. A complementary observation is the failure of the *static* Quantum SVM on these continuous manifolds (Circles: 59.27%; Moons: 63.92%), attributable to periodic feature collisions in its fixed second-order ZZ-feature map. This motivates making the embedding trainable: QKAResQNet replaces the fixed map with a trainable, bias-free alignment feeding a first-order Z encoding. Because the static QSVM and QKAResQNet use different encoding families (ZZ vs. Z), we do not isolate trainability alone; the comparison shows that fixed quantum kernels collapse on this geometry while a trainable hybrid recovers performance. A controlled static-vs-trainable ablation of a single encoding is left to future work.

B. Representational Parity ($p \gg N$ Starvation)

To test stability in a severe feature-starvation regime, Leukemia (7,129 features) and Colon (2,000 features) were evaluated with microscopic sample sizes over 20 random seeds. Table III and Fig. 3 demonstrate representational parity. The 4-qubit quantum bottleneck matches the capacity of the massively parameterized classical bottleneck ($\sim$65%) using a near-identical parameter count, with the aligned hybrids reaching statistical parity (Leukemia: ResQNet $p = 0.471$, QKAResQNet $p = 0.701$; Colon: parity). We use "parity" throughout to mean no statistically significant difference under the reported tests, not proven equivalence. The

TABLE III

EXTREME STARVATION (20-SEED): REPRESENTATIONAL PARITY

Dataset	Model	Accuracy (%)	Total Params
Colon	Classical ResNet	63.63 $\pm$ 11.4	12,037
	Hybrid ResQNet	**65.13** $\pm$ **8.7**	12,029
	Hybrid QKAResQNet	63.35 $\pm$ 11.1	12,029
Leukemia	Classical ResNet	**71.42** $\pm$ 11.2	42,811
	Hybrid ResQNet	68.91 $\pm$ **10.0**	42,803
	Hybrid QKAResQNet	69.55 $\pm$ **9.1**	42,803

unitary constraints act as an implicit regularizer against the chaotic $p \gg N$ noise [4], yielding tighter standard deviations (Leukemia: $\pm$9.1% for QKAResQNet vs $\pm$11.2% classical). Architecture selection is decisive in this regime. Whereas the aligned residual hybrids reach parity, the locality-biased *SplitAttentionQNN* causes statistically significant *harm* on the same genomic data (Leukemia -6.40pp, $p = 0.0016$; Colon -10.24pp, $p = 0.0098$), because chunking destroys the non-local feature correlations that carry the biological signal. This is a structural prediction independent of the result: the attention bottleneck imposes feature locality that conflicts with the non-local correlation structure of genomic data. Inductive bias misalignment therefore degrades performance rather than merely failing to help, establishing alignment as a necessary condition for utility. This pattern extends to two further high-dimensional datasets. On Brain RNA-seq (7 classes, $p = 23{,}433$, $N = 100$), every architecture is pinned near the imbalanced multi-class floor; the aligned QKAResQNet reaches exact parity (43.11% vs 43.61%, $p = 0.93$) while SplitAttentionQNN collapses to 15.33%, the most extreme manifestation of the misalignment failure. On the Indian Pines hyperspectral subset ($p = 200$), all pairs reach parity and the Quantum SVM nearly matches the RBF-SVM reference (71.97% vs 72.28%), because the spectrally smooth features do not trigger the periodic collisions that break static kernels on geometric manifolds.

C. Standard Parity & The Execution Time Penalty

On standard tabular datasets with sufficient sample density (Breast Cancer, QSAR), the hybrid models reached parity on the ResNet/ResQNet pair (Breast Cancer 89.72% vs 88.79%); note that on the hero CKAResCNet/QKAResQNet pair the quantum twin is 3.36pp lower (88.50% vs 85.14%), so the parity statement is pair-dependent on this dataset. However, this parity exposes a severe practical bottleneck. Classical total training times averaged $\sim$0.01 seconds, whereas the quantum models required 24 to 419 seconds over the full 20-epoch run, a three- to four-order-of-magnitude penalty. In the NISQ era, any geometric utility trades off directly against severe wall-clock time penalties.

D. The Classical Compression Boundary

To evaluate hybrid performance on native quantum states, the architectures were tasked with classifying the multipartite entangled NTangled dataset. While purely quantum litera-

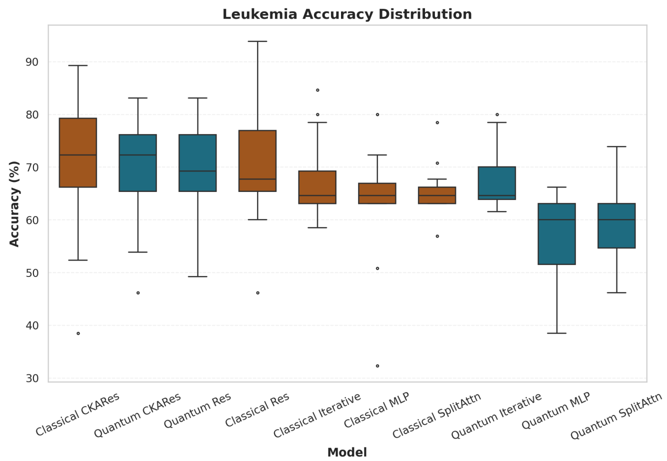

Fig. 3. Leukemia 20-seed accuracy by architecture pair (mean ± std). Accuracies sit near a generalization ceiling of ∼65% due to the microscopic training set. The aligned quantum twins (ResQNet, QKAResQNet) reach statistical parity with tighter variance, while the misaligned SplitAttentionQNN underperforms.

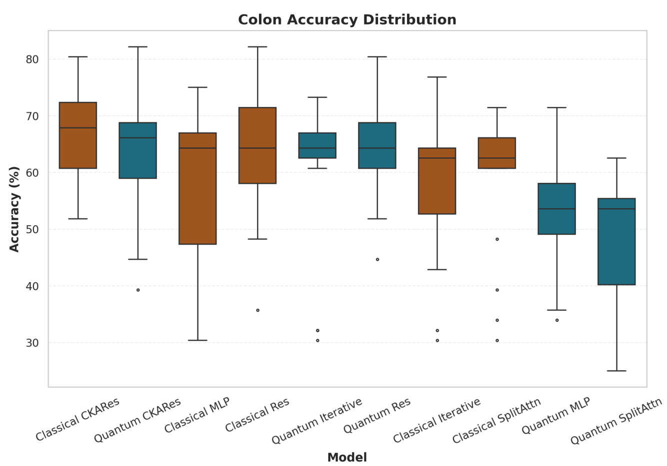

Fig. 4. Colon cancer gene expression 20-seed accuracy by architecture pair (mean ± std). The high-dimensional ($p = 2{,}000$) starvation regime reproduces the Leukemia pattern across a second genomic domain: aligned quantum twins reach parity with their parameter-matched classical twins, supporting the generalizability of the representational-parity boundary.

ture reports >90% accuracy on this dataset, every hybrid architecture evaluated stalled at approximately 50% (random guessing). As illustrated in Fig. 6, the necessary classical dimensionality-reduction layer collapses the high-dimensional entangled input to a low-dimensional real vector, irreversibly discarding the non-local correlations that carry the class signal. This observation is consistent with the structural properties of entangled quantum-native datasets [17], which encode non-classical correlations that classical compression cannot preserve. This empirically defines the *Classical Compression Boundary*: standard classical-to-quantum pipelines that compress the input before encoding are fundamentally unable to process native quantum entanglement.

Fig. 5. The execution-time bottleneck. Total training time (20 epochs) across evaluated datasets on a logarithmic scale. While standard tabular datasets exhibit representational parity, quantum-circuit simulation imposes a severe wall-clock penalty (milliseconds vs. tens to hundreds of seconds).

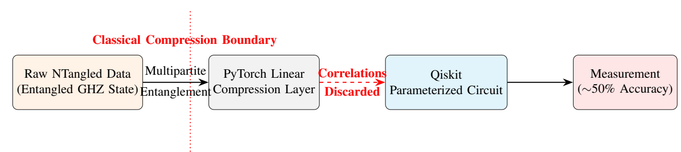

Fig. 6. The Classical Compression Boundary. The classical dimensionality-reduction layer irreversibly discards the non-local correlations of the entangled state before they can be processed by the quantum circuit.

E. Real-World Evaluation Across Hardware Conditions

To move beyond synthetic manifolds and simulation, QKAResQNet was evaluated on real-world ESP32 accelerometer gesture data (10 classes, 56 labeled segments, 96 sensor features) under three conditions (Table IV). The simulation rows use 5-fold cross-validation over the full set; the physical-hardware row uses a single stratified 75/25 split (42 train, 14 test). Under noiseless simulation, QKAResQNet exceeds its classical CKAResCNet twin by +1.81pp (89.39% vs 87.58%) with lower variance, consistent with the synthetic-manifold results. Under the simulated IBM Torino profile (Heron r1, 3.03% readout error), the relationship reverses (−1.82pp), with overlapping variance, indicating that the older device's readout noise erodes the hybrid's edge. On the physical 156-qubit IBM Kingston processor (Heron r2), the model classified 13 of 14 held-out test samples correctly, equal to the RBF-SVM count and one sample above the classical twin (12/14). We report raw counts rather than percentages: at $N = 14$ the differences between hardware, simulator, and twin are all within single-sample sampling noise, so this run is a feasibility check of end-to-end execution, not a powered comparison. The γ-gated residual design is the mechanism behind this hardware resilience: the classical highway carries the bulk of the prediction, so transient quantum errors degrade the output gracefully. The evaluation batch required 84 seconds of wall-clock time, dominated by CPU-QPU input/output latency rather than gate execution, identifying classical-quantum communication overhead as the principal engineering bottleneck

TABLE IV
GESTURE CLASSIFICATION ACROSS NOISE AND HARDWARE CONDITIONS

Condition	Classical	Quantum
Noiseless simulation (5-fold CV)	87.58 ($\pm$7.1)	**89.39 ($\pm$6.5)**
IBM Torino noise (5-fold CV)	**89.55 ($\pm$9.9)**	87.73 ($\pm$10.3)
IBM Kingston (hardware, 14-sample test)	12/14	**13/14**
RBF-SVM (reference)	13/14 hardware test; 91.21 (sim, 5-fold CV)	

for present-day deployment. As a directional real-world signal (single 75/25 split, $N = 14$ test), this awaits larger labeled datasets for statistical confirmation.

F. Limitations

Several constraints bound the scope of these claims. (i) All quantum models use 4 qubits, chosen so that $\sim$2,000 parameter-shift trainings remain exactly simulable and reproducible; the conclusions are not extrapolated to larger registers. (ii) The decision-boundary and gate-trajectory figures are single-seed illustrations of mechanism; the quantitative claims rest on the 20-seed tables. (iii) The physical-hardware result is a single 14-sample test split and is reported as a feasibility check, not a powered comparison. (iv) The $\sim$2,000 trainings report per-regime p-values to delineate operational boundaries; we do not pool them into a single family-wise hypothesis and apply no family-wise correction, so the p-values should be read descriptively. (v) "Parity" denotes no statistically significant difference under the reported tests, not statistical equivalence, which the present seed counts cannot establish. (vi) The RBF-SVM is the reference ceiling only in the regimes where it applies; on the higher-dimensional QSAR and Colon sets a neural twin slightly exceeds it, and it was not run on quantum-native data.

IV. CONCLUSION

This study identifies four operational boundaries for hybrid neural networks. The results do not support a universal quantum advantage; instead, quantum utility is strictly bound to inductive bias alignment between the parameterized circuit and the data manifold. On complex geometries under starvation, quantum feature maps offer measurable topological utility, with QKAResQNet exceeding its classical twin by $+14.42$pp and the best-aligned pair by $+24.48$pp, though on the geometric manifolds all hybrids remain below the exact RBF-SVM, which is the reference ceiling in that regime. (On the high-dimensional QSAR and Colon sets a neural twin slightly exceeds the RBF-SVM, and on quantum-native data the RBF-SVM is not applicable, so the ceiling claim is scoped to the geometric regime.) On extreme $p \gg N$ genomics, a 4-qubit bottleneck matches the representational capacity of massive classical networks with tighter variance, while a misaligned architecture (SplitAttentionQNN) causes statistically significant harm. These benefits are constrained by severe NISQ execution-time penalties and by the classical compression of native quantum states. Physical deployment on IBM Kingston confirms that QKAResQNet executes end-to-end on real hardware (13/14 on a small feasibility sample), with classical-quantum I/O latency, not circuit execution, as the dominant practical barrier. Future work will expand physical-hardware deployment, add adequately powered real-world datasets, and run a controlled static-vs-trainable encoding ablation to convert these directional results into statistically confirmed claims.

REFERENCES

[1] H.-Y. Huang, M. Broughton, M. Mohseni, R. Babbush, S. Boixo, H. Neven, and J. R. McClean. Power of data in quantum machine learning. *Nature Communications*, 12:2631, 2021.
[2] M. C. Caro, H.-Y. Huang, M. Cerezo, K. Sharma, A. Sornborger, L. Cincio, and P. J. Coles. Generalization in quantum machine learning from few training data. *Nature Communications*, 13:4919, 2022.
[3] M. C. Caro, E. Gil-Fuster, J. J. Meyer, J. Eisert, and R. Sweke. Understanding quantum machine learning also requires rethinking generalization. *Nature Communications*, 2024.
[4] L. Banchi, J. Pereira, and S. Pirandola. Generalization in quantum machine learning: a quantum information standpoint. *PRX Quantum*, 2:040321, 2021.
[5] M. Cerezo, A. Arrasmith, R. Babbush, S. C. Benjamin, S. Endo, K. Fujii, and P. J. Coles. Variational quantum algorithms. *Nature Reviews Physics*, 3(9):625–644, 2021.
[6] M. Schuld, A. Bocharov, K. M. Svore, and N. Wiebe. Circuit-centric quantum classifiers. *Physical Review A*, 101(3):032308, 2020.
[7] M. Schuld, R. Sweke, and J. J. Meyer. Effect of data encoding on the expressive power of variational quantum machine-learning models. *Physical Review A*, 103:032430, 2021.
[8] S. Thanasilp, S. Wang, M. Cerezo, and Z. Holmes. Exponential concentration in quantum kernel methods. *Nature Communications*, 15:5200, 2024.
[9] C. Blank, D. K. Park, J.-K. K. Rhee, and F. Petruccione. Quantum classifier with tailored quantum kernel. *npj Quantum Information*, 6:41, 2020.
[10] J. R. McClean, S. Boixo, V. N. Smelyanskiy, R. Babbush, and H. Neven. Barren plateaus in quantum neural network training landscapes. *Nature Communications*, 9(1):4812, 2018.
[11] N. A. Nemkov, E. O. Kiktenko, and A. K. Fedorov. Barren plateaus swamped with traps. *Physical Review A*, 111(1):012441, 2025.
[12] C. Ortiz Marrero, M. Kieferová, and N. Wiebe. Entanglement-induced barren plateaus. *PRX Quantum*, 2:040316, 2021.
[13] A. K. Fedorov, N. Gisin, S. M. Beloussov, and A. I. Lvovsky. Quantum computing at the quantum advantage threshold: a down-to-business review. *arXiv preprint arXiv:2203.17181*, 2022.
[14] J. Bowles, S. Ahmed, and M. Schuld. Better than classical? The subtle art of benchmarking quantum machine learning. *arXiv preprint arXiv:2403.07059*, 2024.
[15] D. Basilewitsch, J. F. Bravo, C. Tutschku, and F. Struckmeier. Quantum neural networks in practice: a comparative study with classical models from standard data sets to industrial images. *Quantum Machine Intelligence*, 7(2):110, 2025.
[16] A. Liu, C. Wen, and J. Wang. Lean classical-quantum hybrid neural network model for image classification. *Advanced Quantum Technologies*, 2400703, 2025.
[17] L. Schatzki, A. Arrasmith, P. J. Coles, and M. Cerezo. Entangled datasets for quantum machine learning. *arXiv preprint arXiv:2109.03400*, 2021.
[18] U. Shrikant and P. Mandayam. Quantum non-Markovianity: Overview and recent developments. *Frontiers in Quantum Science and Technology*, 2:1134583, 2023.